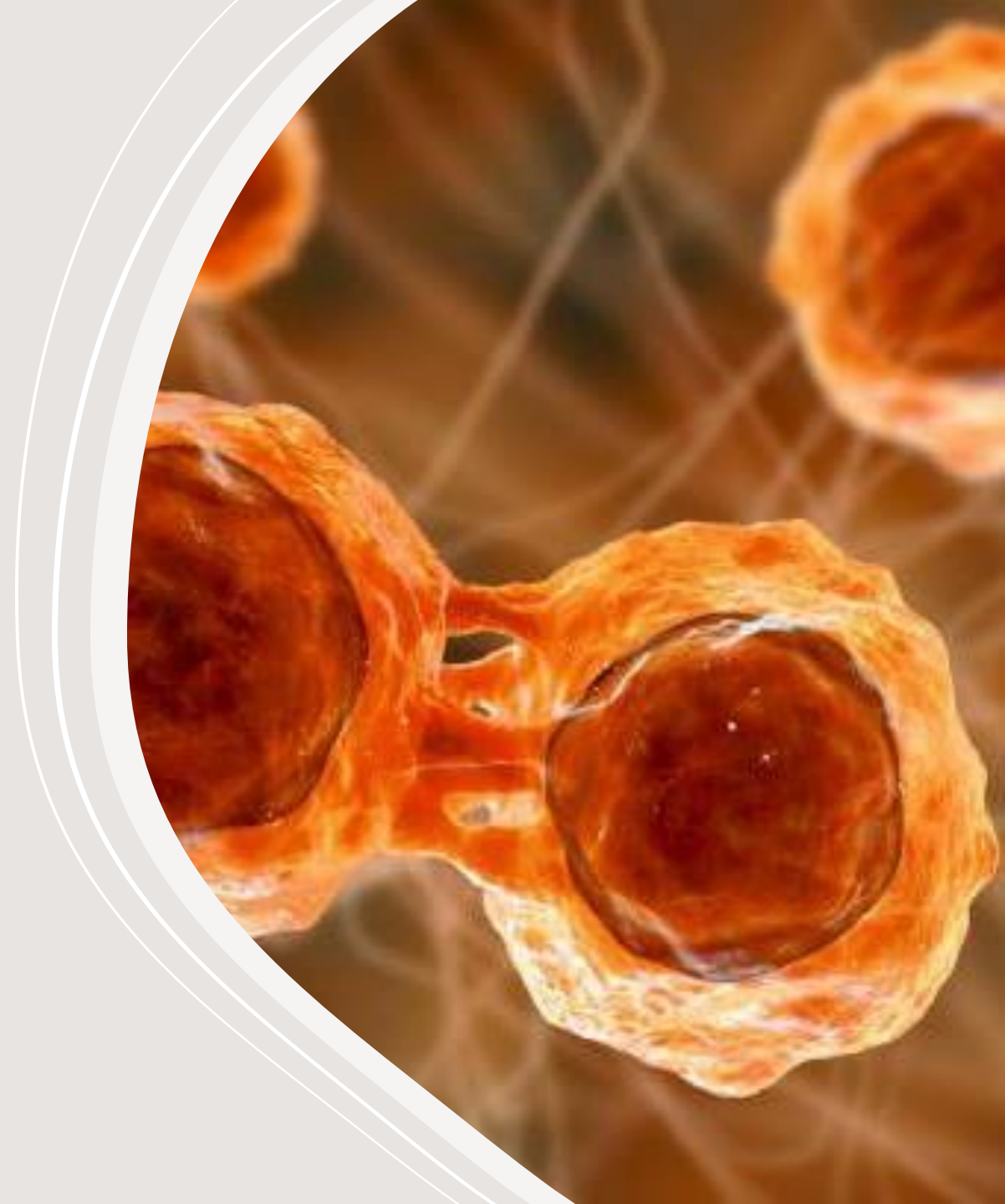


Myeloma



- 
- Myeloma cells : neoplasm cells (cancer cells) Antibody secreting cells.

Neoplasm: cells with uncontrolled division and Abnormal structure.

- Myeloma cells secrete Antibodies Abnormal Antibodies.
- Myeloma cells are type of cancer (cancer B cells) which mean can divide indefinitely (immortal).

- 
- Normally when the Ag enter the body, specific B cells will recognize this Ag, and Proliferate to produce Clones of B cells recognize the Ag, some B cells become memory B cells and some become plasma cells.
 - Plasma cells: the cells that secrete Antibodies.
 - Plasma cells have short life span then die.
 - Plasma cells(Ab secretion) + myeloma cells(immortal cell) → Hybridomas (cells can secrete Abs and immortal)



- Myeloma cells use in hybridoma technology are :

- Immortal

- HGPRT- (Hypoxanthine Guanine phosphoribosyl Transferase)

Nonproducing and nonsecreting Abs

- HGPRT: an enzyme play key role in Salvage pathway.
- HGPRT present on single X active chromosomes, point mutation in gene lead to enzyme inactivation.
- Then incubate the cells with 8-Azaguanine or 6-thioguanine.
 - Aza Guanine and Thio Guanine are analogs of Guanine and toxic to Cells.
- Cells with HGPRT – (enzyme inactive) can't incorporate Azaguanine or Thio Guanine in host DNA • cell survive.
- While myeloma cells that have active HGPRT (HGPRT+) will incorporate these toxic analogs in Myeloma DNA which lead to Cell die.



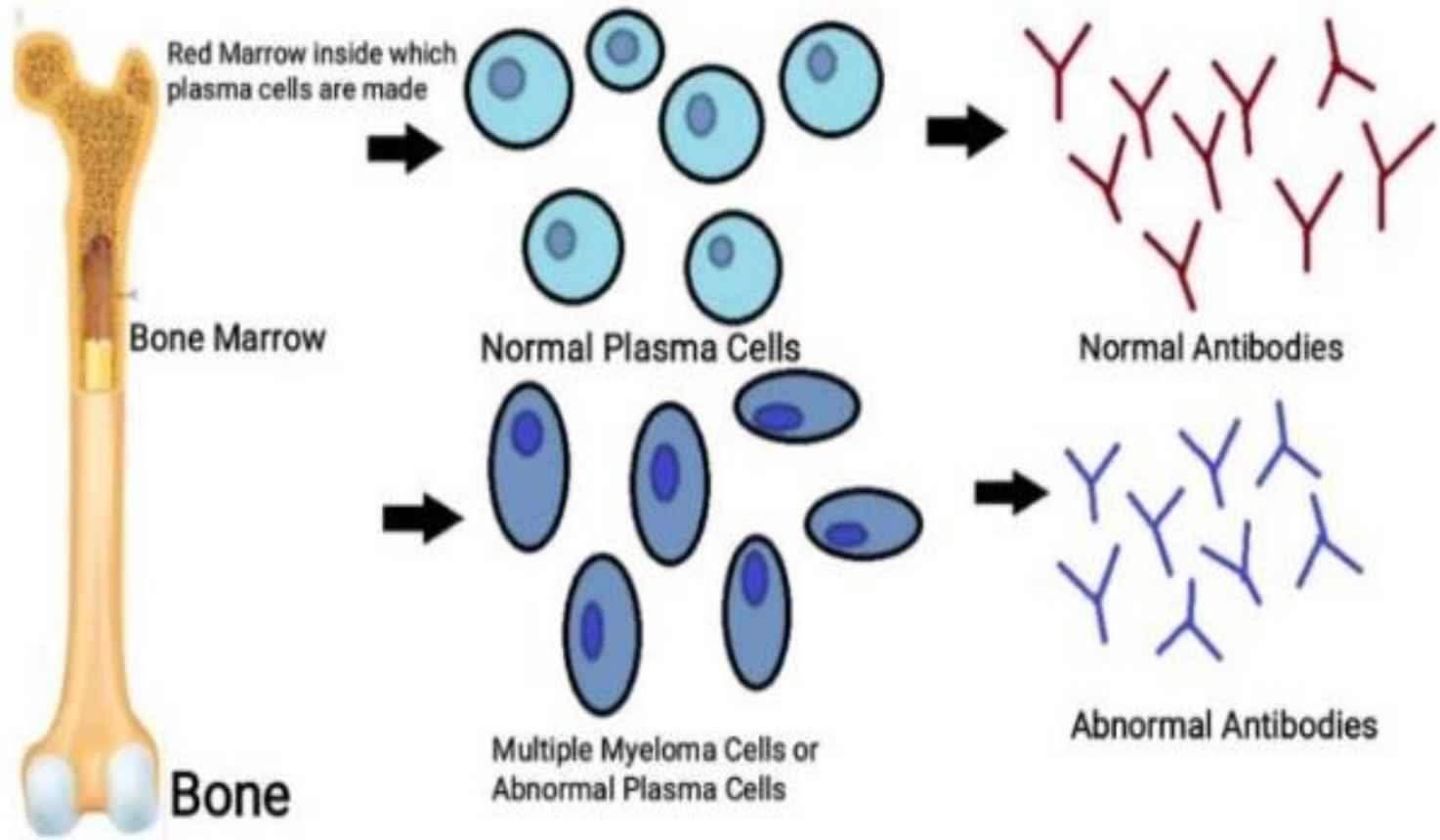
- Types of myeloma cells used in Hybridoma technology :

From Balb /C

- Sp2/0-Ag14

- SP2

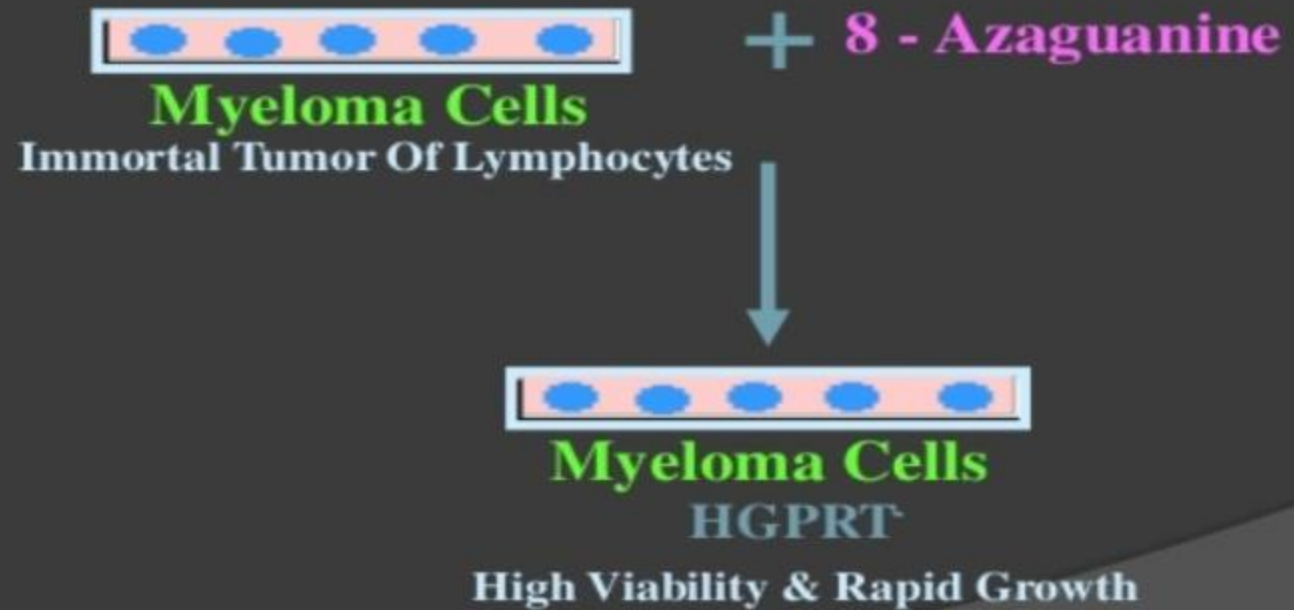
- NS-1



PRODUCTION OF MONOCLONAL ANTIBODY

HYBRIDOMA TECHNOLOGY

Step 3: - Preparation of Myeloma Cells



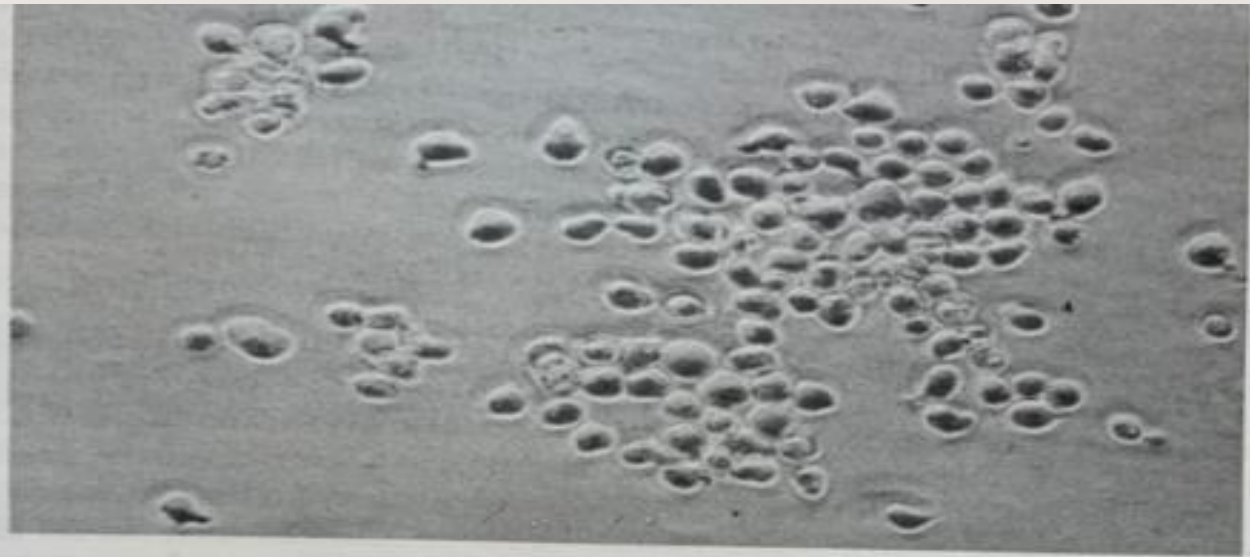


Fig. 5.1.3a Mouse myeloma cells (P3-X63-Ag8.653)



Fig. 5.1.3b Rat myeloma cells (YB2/3.0 Ag20)



- Using cells from human for monoclonal production is hampered by :

1. Source of Highly sensitized immune lymphocytes.

2. Cant inject a human with foreign Ag.

3. Human myeloma are difficult to establish in tissue culture

- 
- Human myeloma have Abundant RER, mitochondria and Golgi associated with high Ig production.
 - Human Myeloma Don't carry EBV and are aneuploidy
 - Doubling time in culture 70 hrs.

- **Maintain of Myeloma in tissue culture :**

- ★ Cells are maintained in Tissue culture tube.

- ★ Tube contain 50 ml complete media, 10% FCS, 5% CO₂ and 37C°, Penicillin and Streptomycin.

- ★ Density of cells $10^5 - 5 \times 10^6$ cell/ml.

To little density → slow growth

High much density → crowding and cells death

- ★ When the cells reach 80% confluence cells must be passage and add fresh media.

- ★ Doubling time 12-18 hrs



★ Myeloma cells that stick can be easily resuspended by gentle scraping by a syringe and kwill or plastic Pasteur, but some human lines need to be detached by trypsinization.

★ Medium should never be poured from the flask • use the Pipettes.

★ Medium should never be poured from the flask

★ It is inadvisable to maintain myeloma cells in culture indefinitely since they are prone to spontaneous mutations and revert to HGPRT+. The cells should be grown periodically in media containing 6-TG or 8-AG. HGPRT- Will be unaffected but any HGPRT+ die.

★ Myeloma should be storage in Liquid nitrogen.

Counting myeloma cells :

★ Using haemocytometer slide or cell counter

★ Count all the cells in each of the 4 large corner squares. The total is divided by 4 to obtain the mean number

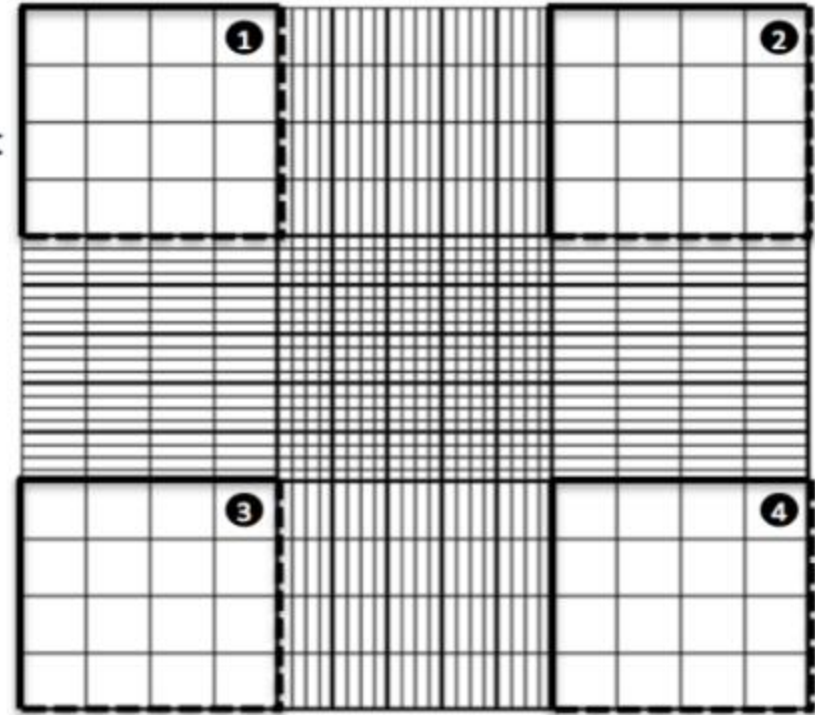
★ The average = $53 + 51 + 47 + 49 / 4 = 50$

The total volume of each square: $1 \text{ mm} \times 1 \text{ mm} \times 0.1 \text{ mm} = 0.1 \text{ mm}^3 = 0.0001 \text{ cm}^3 = 10^{-4} \text{ ml}$

Multiply by 10^4

★ Cell No in 1 ml = $50 \times 10^4 = 5 \times 10^5 \text{ cell/ml}$

— count
--- don't count



Viability test

- ★ Ethidium bromide or Acridine orange

- ★ Using haemocytometer,

No preincubation is necessary, counting under fluorescent microscope

- ★ Ethidium bromide will be taken up by dead cells nuclei fluoresce

- ★ Acridine orange will be taken up by viable cells, and their nuclei fluoresce green.

- ★ Both dyes are mutagenic and should be handled with care

Trypan blue stain

- Taken by dead cells immediately, should be read within 1-2 min.

Disadvantage:

- Rapidly taken by viable cells within 10 min
- Have great affinity to proteins so elimination of serum

from the cell diluent would give greater accuracy

- mutagenic and should be handled with care

Storage of cells

- Density: $2-10^6$ cells in 0.25 ml freezing medium/cryotube
- Freezing medium Glycerol or 4%DMSO +96%FCS
- This step should be carried out quickly to minimize DMSO contact with cells at room Temp
- Place in polystyrene box in -80°C immediately and transfer to Liquid N_2 within 24 h
- Loss of specificity has been known to occur

Thawing

- Thaw cryotube as quickly as possible in 37°C water bath
- Dilute gradually with cold media
- Centrifuge and suspend pellet in warm thymocyte media and transfer to 24-well plate. Thymocyte medium is not always necessary but it is better to be safe than sorry.